

for formatting C programs. Astyle also has other options to set parameters like the bracket style, tab size, indentation style, empty line padding, etc. Figure 2 shows a program called *astyle_example.c* formatted in a style that ought to be called 'chaos'. If the program was a bit lengthier, it would have been impossible to fathom and any decent programmer would curse me for my insolence. But the figure also displays the magic of Astyle by changing the format of the program to GNU style. Yes, the ugly duckling has finally become a swan. The original unformatted program is stored in a file called *astyle_example.c.orig*.

Document generators

While writing efficient programs involves logical reasoning and is hence a task for the left hemisphere of the human brain, preparing documentation involves creativity and is handled by the right hemisphere of the brain. No computer in the world can perform creative writing, but automatic document preparation from specially commented programs has become a reality. Document generators are able to produce API documentation for programmers and user manuals for end users. Doxygen is one of the most widely used document generators and one that is highly suitable for the C language. It can cross-reference documentation and code, and the documentation part is written within the code. The program fragment shown below contains Doxygen based comments. The comments are ignored by the C compiler but they are recognised and processed by Doxygen.

```
/**
 * @file doxygen_example.c
 * @author deepu
 * @date 15 Jul 2014
 * @brief Example for using doxygen with C.
 */
```

Doxygen can be called as follows to process the C file: *doxygen program_name.c*. It can prepare documents in the form of HTML, XML, RTF, etc. Doxygen can also produce a LaTeX file as output, which can be further processed to obtain output in the PDF or DVI format. The example program given above uses only the basic tags. There is a large set of tags available to prepare all kinds of documentation.

Open source libraries

One severe criticism that C has faced over the years is with respect to the scarcity of useful library functions. To some extent, this is true, because when compared with languages like Java or Python, the number of library functions is comparatively less. But to overcome this drawback and to aid us in development, there are countless open source libraries that can be used along with the C standard library. The open source libraries range from the ones which allow us to draw images, to the ones which add security to our systems.

Some of the open source libraries widely used along with the C language include OpenGL, OpenCV, OpenSSL,

etc. OpenGL is a cross language application programming interface to handle 2D and 3D vector graphics. It is used in CAD, flight simulation, scientific visualisation, etc. OpenCV is a library of functions that can be used for applications related to computer vision. Even though the latest interface is based on C++, OpenCV still maintains an older C-based interface. OpenSSL is an implementation of SSL and TLS. It contains a lot of cryptographic functions to be used in security related applications. And remember, this is just the tip of the iceberg; there are hundreds of useful open source libraries.

A thorough discussion of all the tools and packages described in this article will take up scores of text books. I hope this introduction will help you start a lifelong journey on interacting with the elegant language called C. The best thing about the tools discussed here is that they are all open source technologies. Hence, there are countless manuals freely available on the Internet. Go through the manuals and learn these tools; you can use them with many other languages and platforms because most of them are not limited to C. 

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